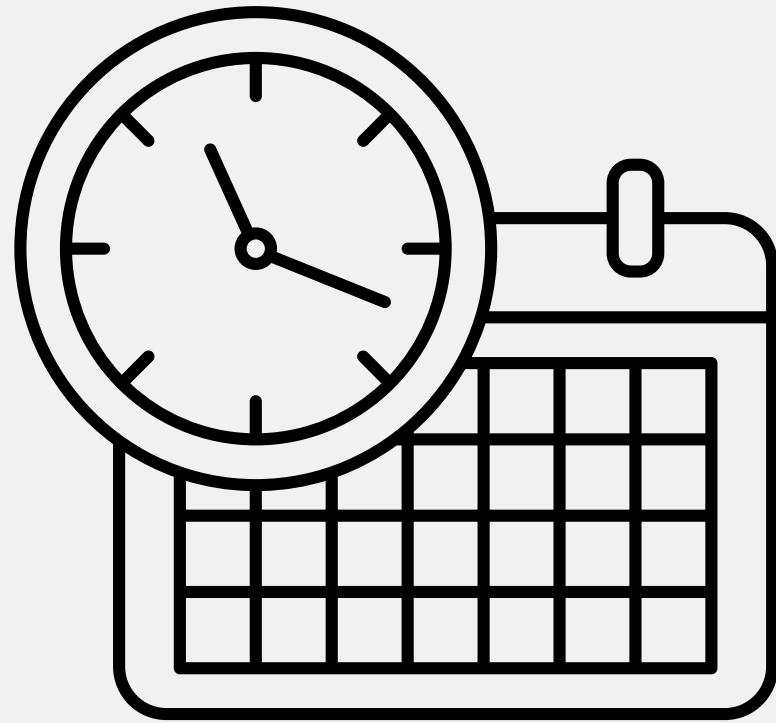


STUDENTS SOCIAL MEDIA ADDICTION

PRESENTED BY: AIDA OUI YOUSRA CHAHINEZ

Introduction



Students spend
hours daily on
social media



It affects focus,
grades & sleep



Causes stress,
anxiety & conflicts

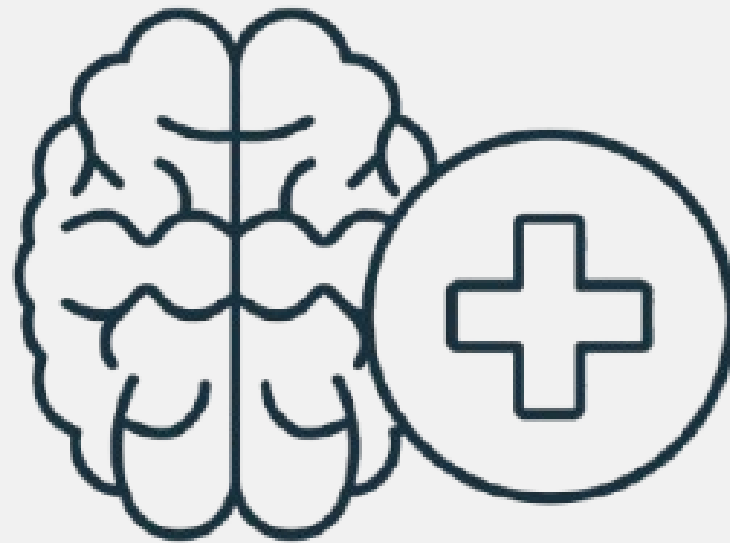
“ARE WE CONTROLLING OUR
SCREENS, OR ARE THEY
CONTROLLING US?”



Using AI to Predict Addiction Scores



Understand what
causes social
media addiction



Explore how sleep,
focus, & mental well-
being affect it



Help schools &
parents build smarter
prevention strategies

Dataset Overview

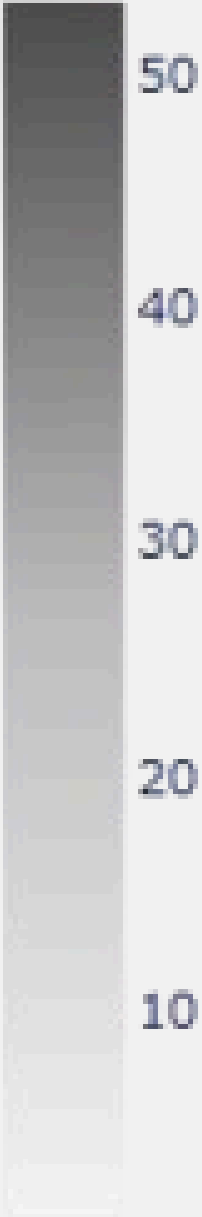
This dataset captures various aspects of students lives, including :

Main Feature Categories	
Demographics	Age, Gender, Country, Academic Level
Behavior	Daily Usage Hours, Most Used Platform
Mental & Emotional	Sleep_Hours_Per_Night , Mental_Health_Score:
Academic & Social	Affects_Academic_Performance , Conflicts_Over_Social_Media ,Relationship_Status

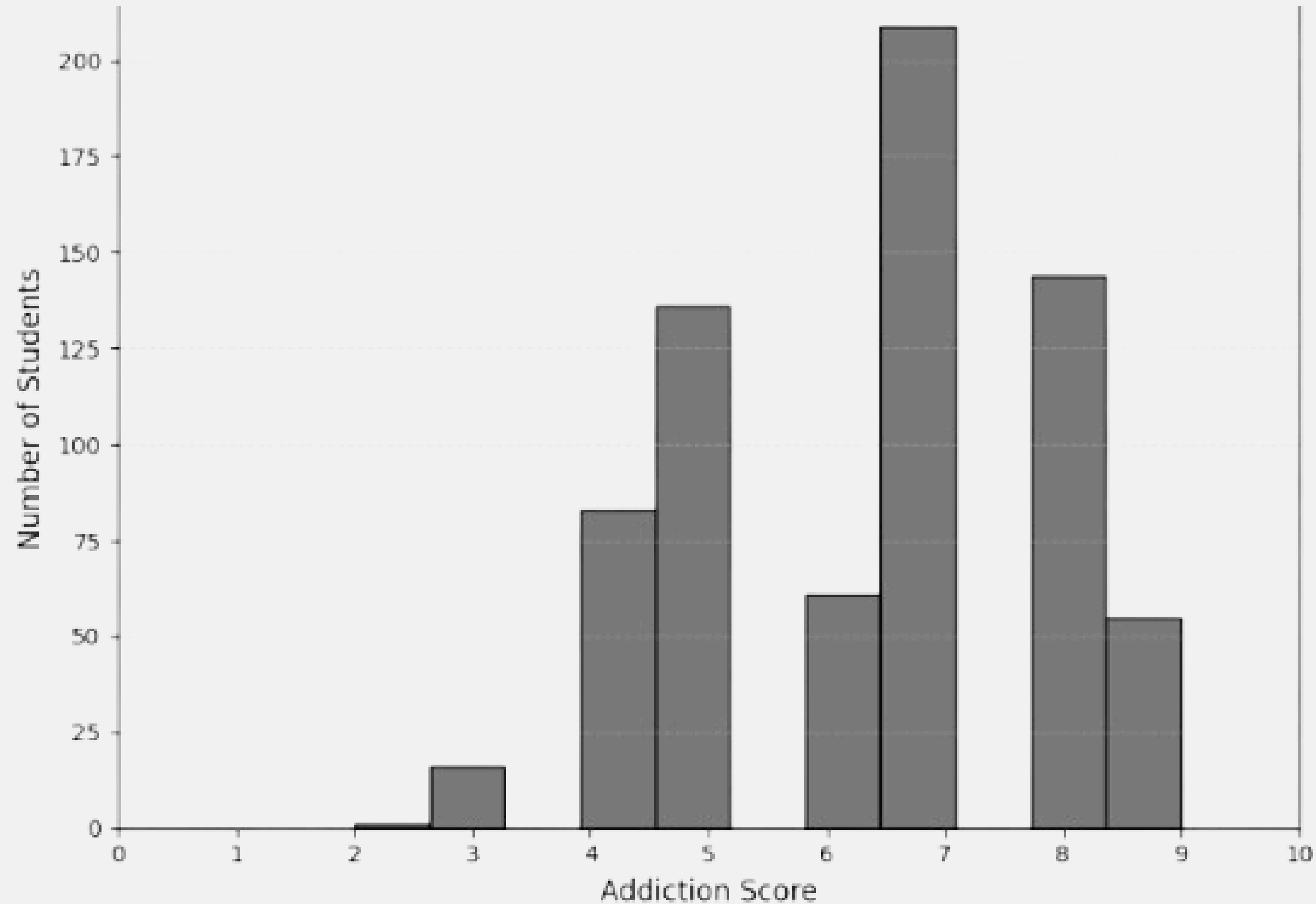
Global Participation



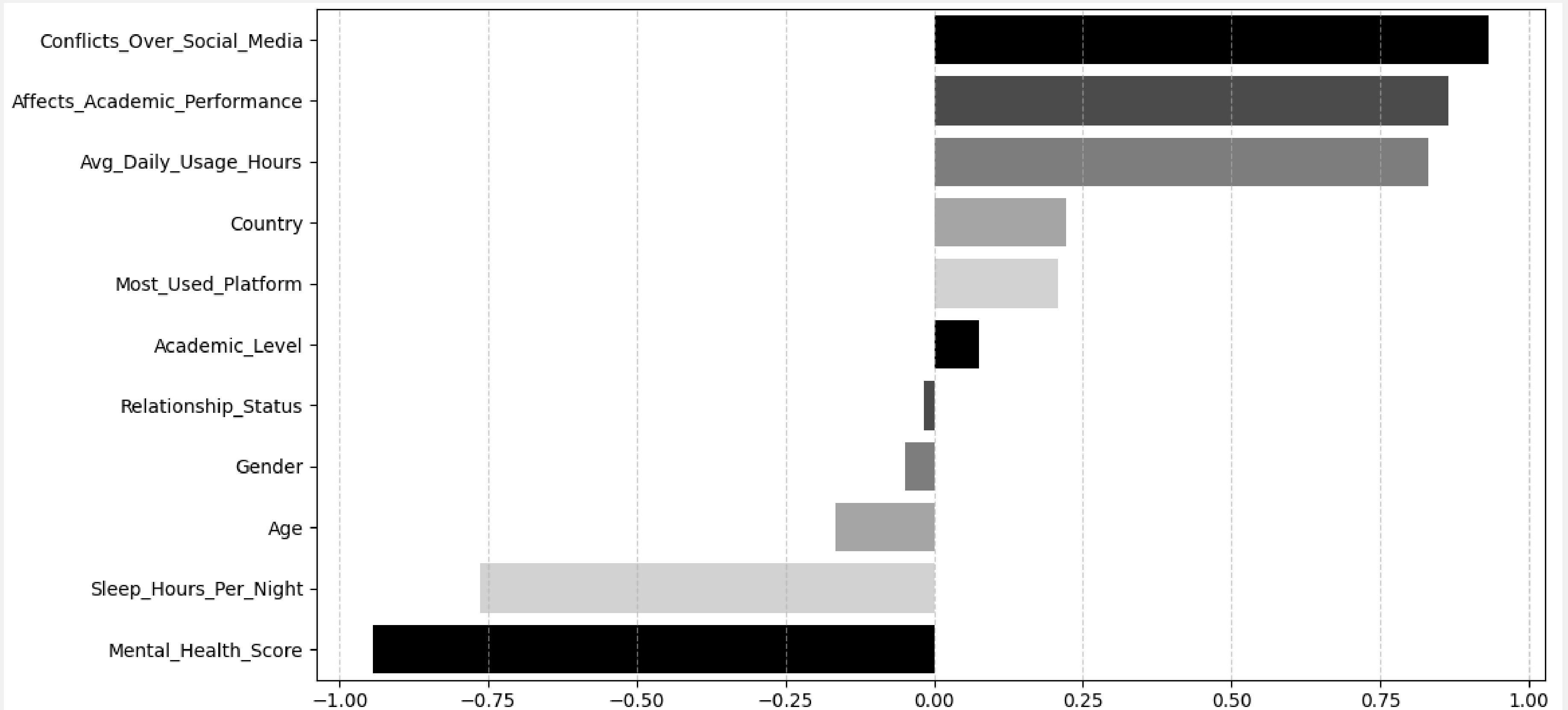
Number_of_Students



Addiction Score Distribution



What Factors Are Linked to Social Media Addiction?



What Factors Are Linked to Social Media Addiction?



More conflicts
over social media



Less sleep



Students who say it
affects academics



Lower mental
health

Predictive Modeling Approaches

Two Approaches are used :

Regression

Predict the exact
Addiction Score

Classification

Classify students
as Low, Moderate,
or High addiction

Approach 1 – Regression: Predicting Addiction Score

MODEL USED

Random Forest Regressor

EVALUATION METRICS

- Mean Squared Error
- R^2 Score

Approach 1 - Predictive Modeling Process

STEPS :

STEP 1

Data Cleaning

STEP 2

Encoding Categorical
Features

STEP 3

Model Selection

STEP 4

Model
Evaluation

Step 1 : Data Cleaning

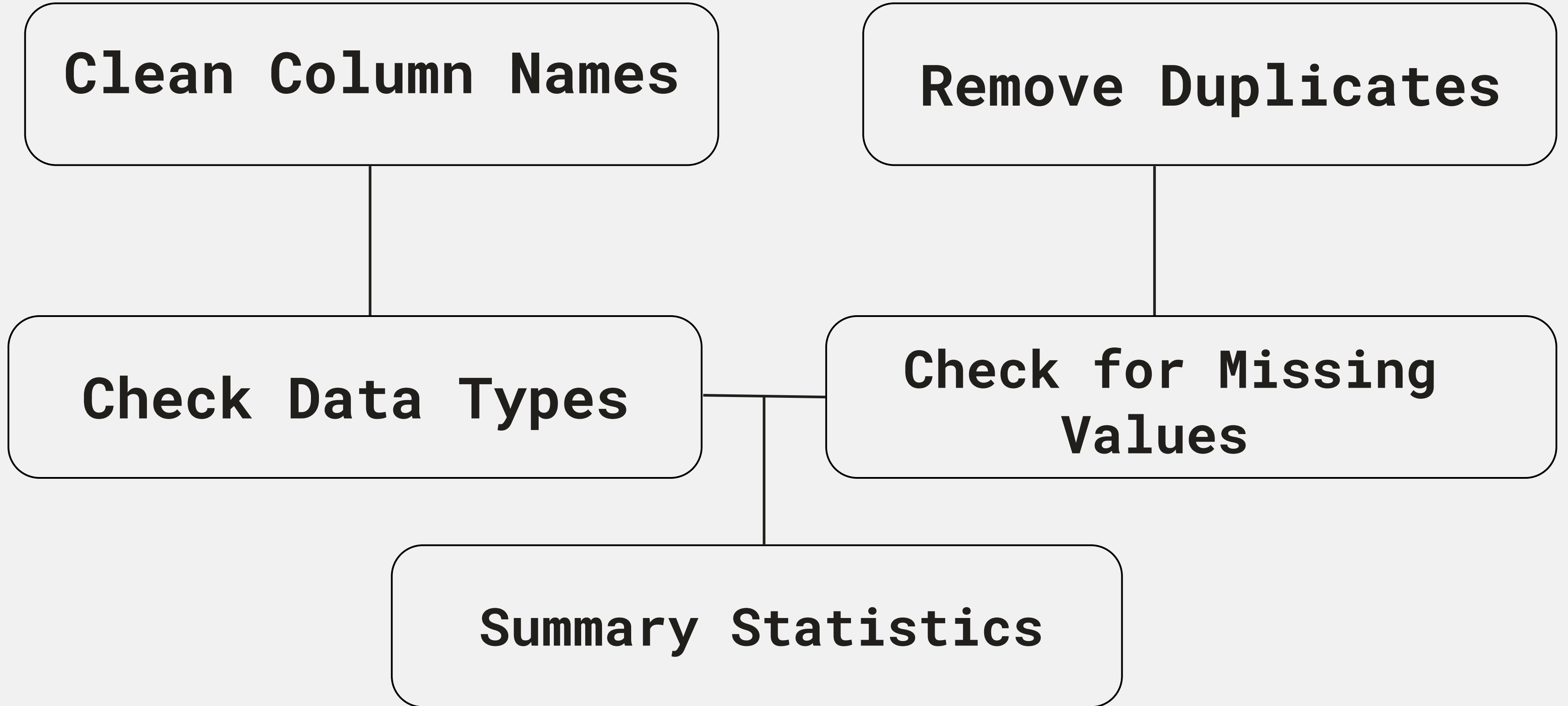
Clean Column Names

Remove Duplicates

Check Data Types

**Check for Missing
Values**

Summary Statistics



Step 2 : Encoding Categorical Features

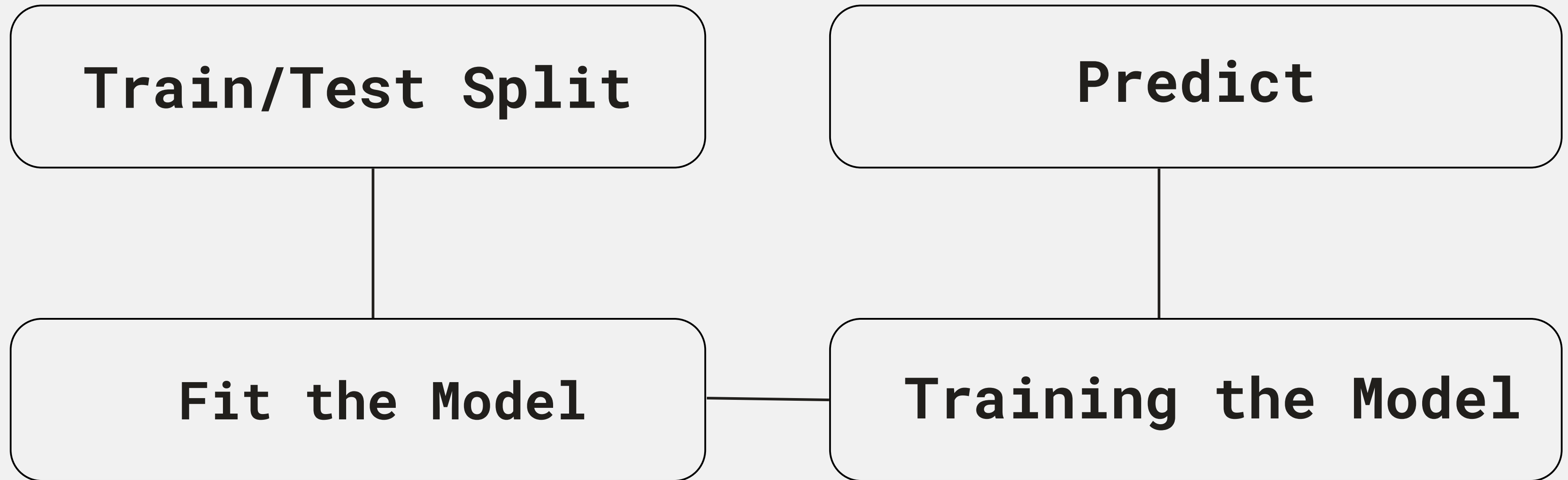
01

CONVERT CATEGORICAL VALUES INTO NUMERIC
FORMAT FOR MACHINE LEARNING MODELS

02

SAVED TO ENCODED_DATASET.CSV

Step 3 : Model Selection



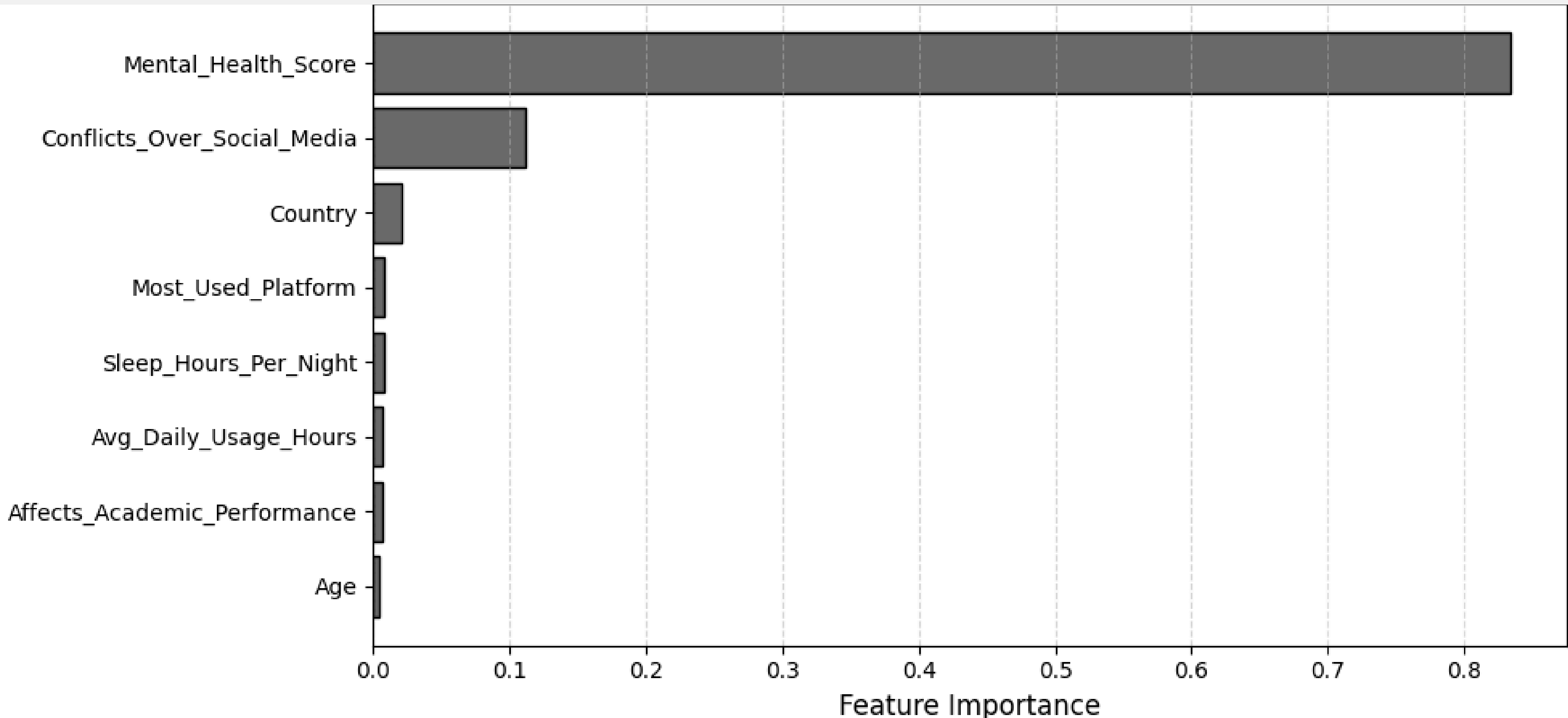
Step 4 : Model Evaluation

Mean Squared Error	R ² Score
0.05	0.98

Indicates a highly accurate and reliable
prediction model

Feature Importance for Predicting Addiction Score :

Identify which features contribute most to the model's predictive accuracy



Approach 2 – Classification: Predicting Addiction Category

MODEL USED

Random Forest Classifier

EVALUATION METRICS

- Accuracy
- Precision & Recall
- f1-score
- Confusion Matrix

Approach 2 - Predictive Modeling Process

STEP 1

Data Cleaning

STEP 3

Label Creation

STEP 5

Model Selection

STEP 2

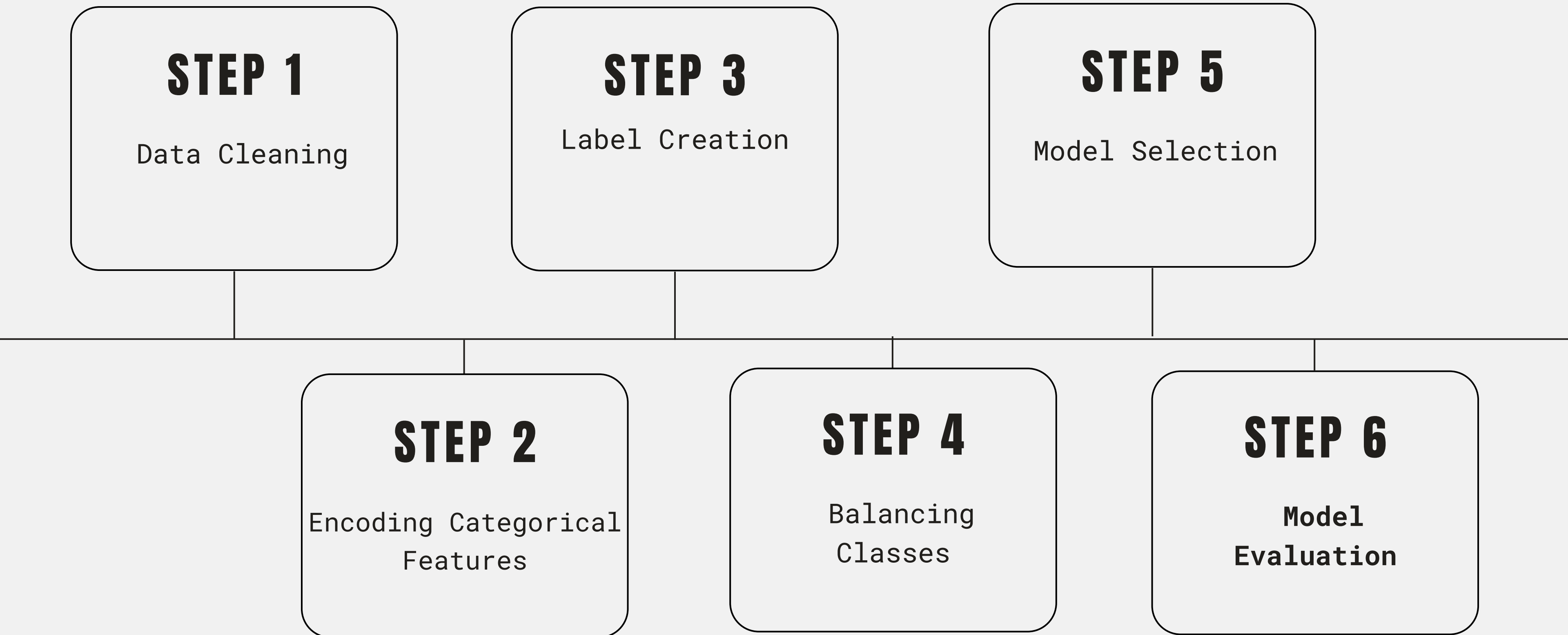
Encoding Categorical
Features

STEP 4

Balancing
Classes

STEP 6

Model
Evaluation



Step 1 : Data Cleaning

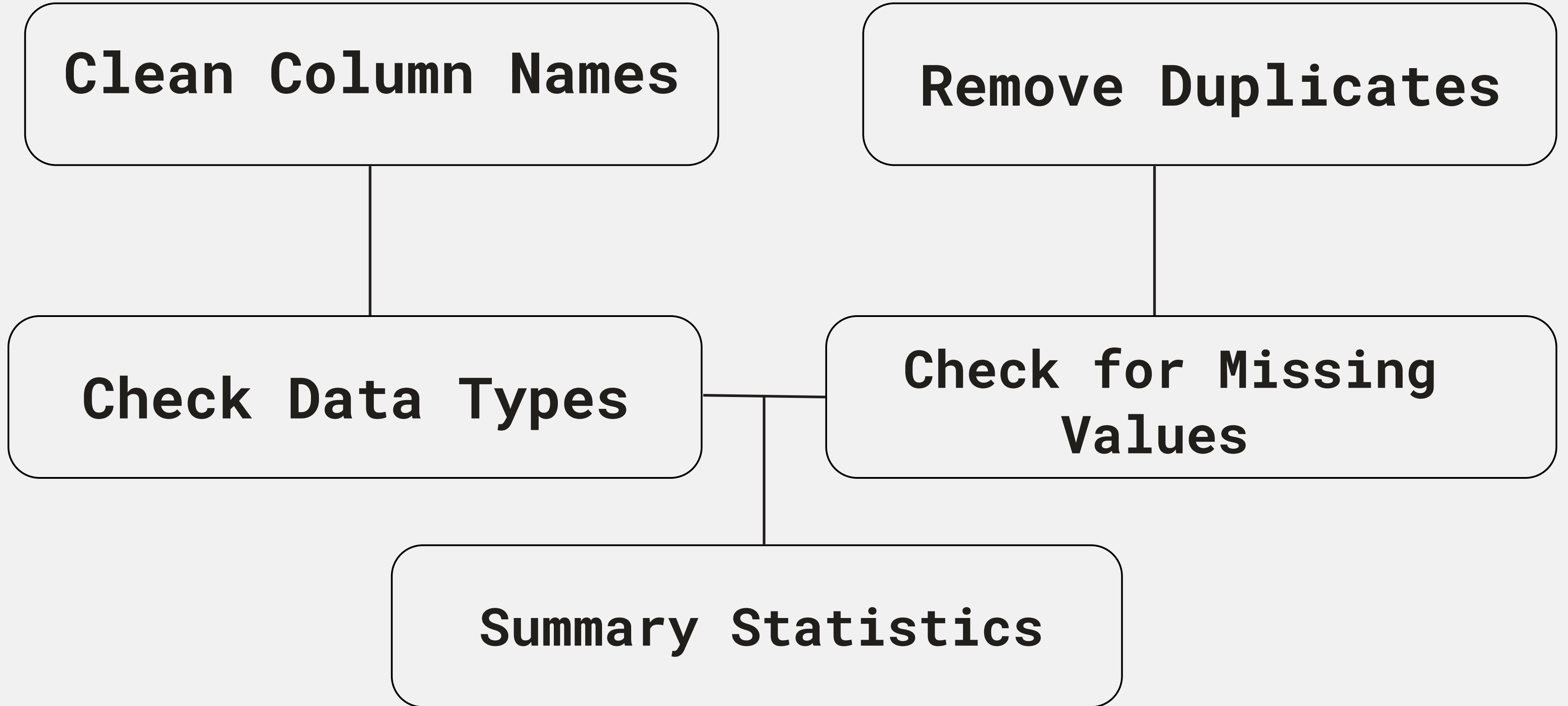
Clean Column Names

Remove Duplicates

Check Data Types

**Check for Missing
Values**

Summary Statistics



Step 2 : Encoding Categorical Features

01

CONVERT CATEGORICAL VALUES INTO NUMERIC
FORMAT FOR MACHINE LEARNING MODELS

02

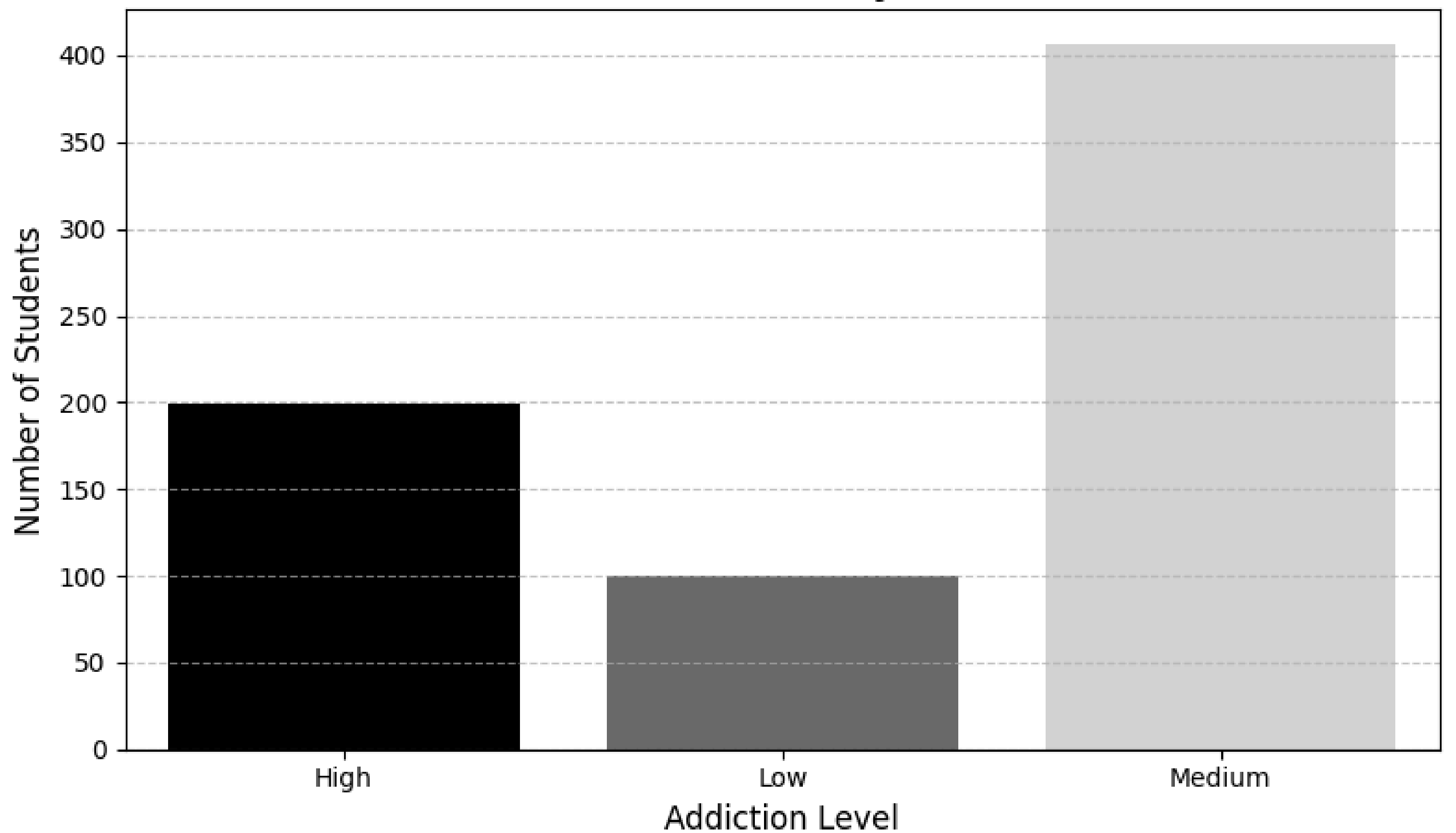
SAVED TO ENCODED_DATASET.CSV

Step 3 : Label Creation

Classify students into Low, Medium, or High addiction levels

Addicted Score	Addiction Level
0-4	Low
5-7	Medium
8-10	High

DISTRIBUTION OF ADDICTION LEVELS



Step 4 : Balancing Classes

Why Balance the Classes?

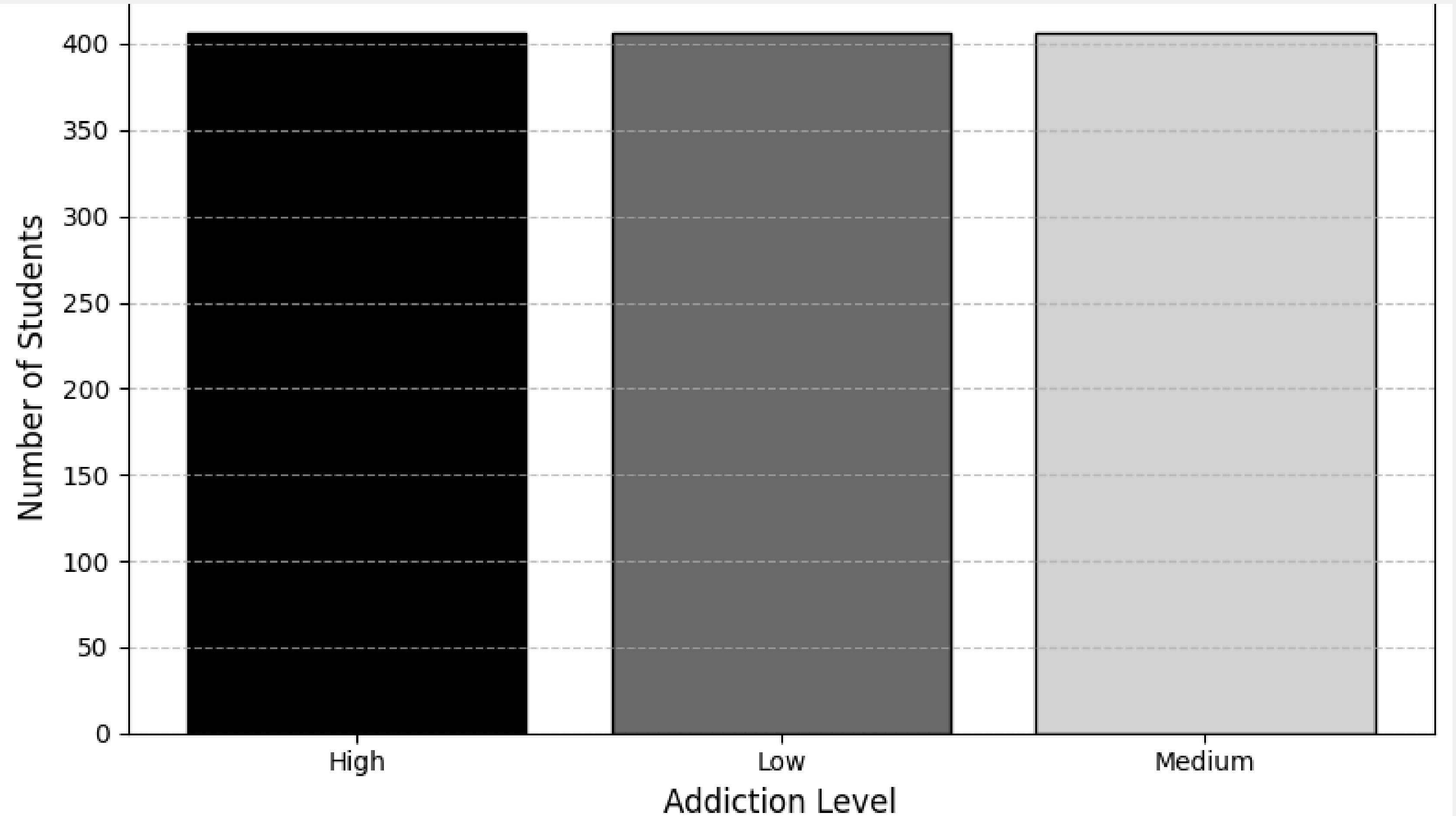
The original dataset had imbalanced classes: "Low", "Medium", and "High".

This could lead the model to favor the majority class, reducing overall performance.

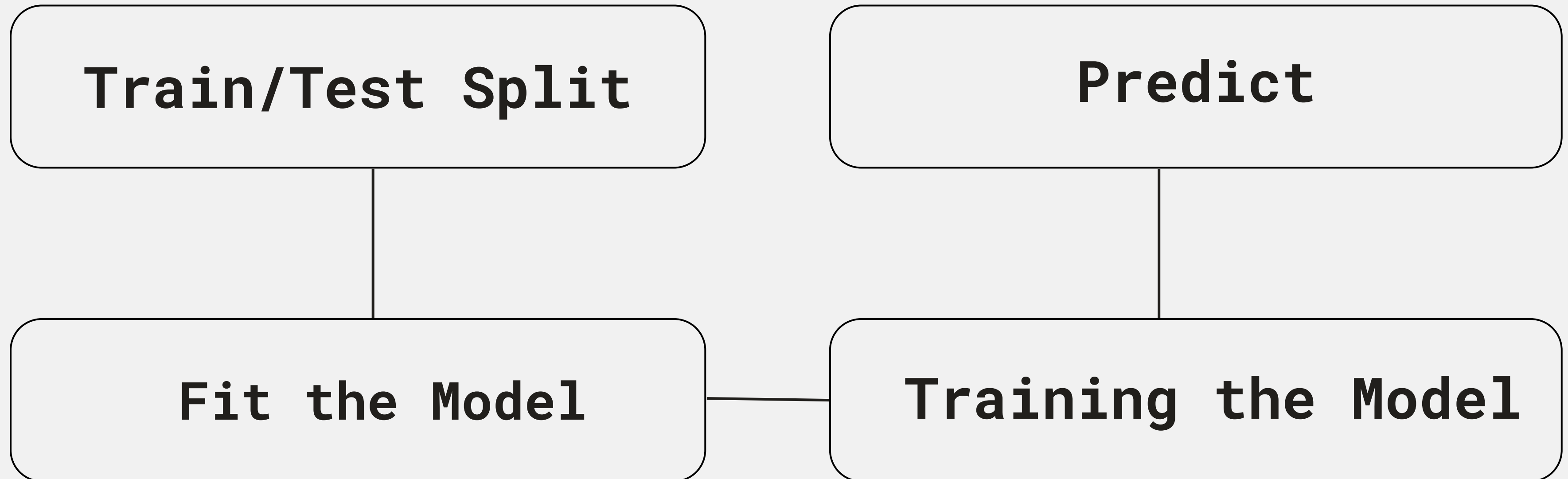
Method Used:

SMOTE (Synthetic Minority Over-sampling Technique): generates synthetic samples for the minority classes to ensure the model learns equally from all classes

CLASS BALANCE AFTER SMOTE



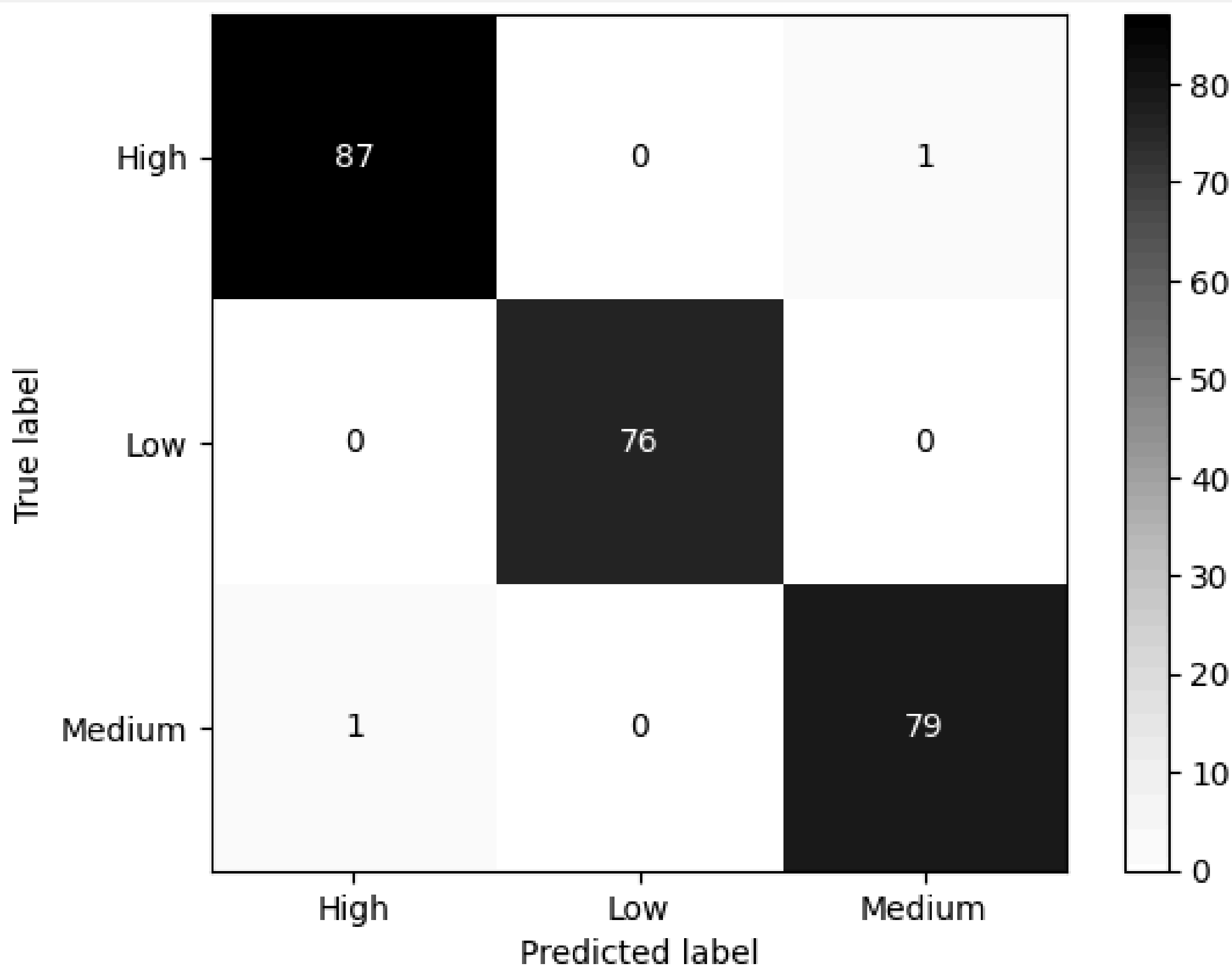
Step 5 : Model Selection



Step 6 : Model Evaluation

Class	Precision	Recall	F1-score
High	0.99	0.99	0.99
Low	1.00	1.00	1.00
Medium	0.99	0.99	0.99
Overall Accuracy	0.99		

CONFUSION MATRIX



Conclusion

Social media addiction is a growing concern among students, impacting mental health and academic performance.

By leveraging AI models we were able to **detect** and **classify addiction levels**.

This approach can help **educational institutions** and **mental health professionals** identify **at-risk students** early and **promote** healthier digital habits.



THANK YOU



Access the Full Code on Google Colab:

<https://colab.research.google.com/drive/1cdrWY-0QdDM4De7xpm423BDCMnLUWMYa?usp=sharing>